



Al-imam Mohammed bin Saud Islamic University
College of Computer and Information Sciences
Computer Science Department

2024/1446

CS290 Project: Tourist Guide System

(phase 1)

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section: 171

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1. Project Initiation:

1.1 Project Scope:

- The Tourist Guide System is designed to be an all-in-one resource for travelers, offering an easy-to-use platform that helps them discover and navigate destinations in Saudi Arabia. This application will store and present detailed information on tourist attractions, lodging, transport options, and events, enabling quick access to necessary details for a seamless travel experience.

1.2. Objectives:

- Develop a user-centric platform that provides comprehensive and easy access to information on tourist sites and services.
- Allow users to create personalized travel plans based on their interests, time, and preferences, ensuring a tailored travel experience.
- Provide up-to-date information on events, weather, and transport schedules to assist travelers in making real-time decisions.
- Incorporate robust data security measures to protect user information and adhere to privacy laws.
- Ensure a consistent and reliable experience across various platforms and devices, including offline functionality for key features.

1.3. Target Audience:

The primary users include:

- International and local tourists who need a dependable tool for planning trips and exploring destinations in Saudi Arabia.
- Individuals looking to find and book accommodations, plan their transport, and get real-time information about the places they are visiting.
- Service providers like hotels and tour operators who want to connect with potential customers through the platform.

1.4. Vision for System Features and Capabilities:

The system's vision is to offer a comprehensive platform that not only meets current traveler needs but also adapts over time based on user feedback and new technologies. The system aims to:

- Serve as a central hub for accurate and comprehensive tourist information in Saudi Arabia.
- Feature interactive maps and GPS for easy navigation and orientation in unfamiliar areas.
- Provide a seamless booking experience for various accommodations and transportation modes.
- Include strong data protection and privacy measures to safeguard user data.
- Offer multi-platform compatibility, allowing users to access essential features even without an internet connection.

2. project backlog:

2.1. Product Backlog Description:

Features	Feature Description	Priority
Destination Information	offering detailed insights into top tourist attractions, historical sites, museums, and natural wonders in Saudi Arabia for a comprehensive travel guide.	High
Itinerary Planning	Providing tools to help users create personalized travel plans based on their preferences, schedule, and interests for a unique, easy, and organized trip.	High
GPS and Real-time Updates	Users can be informed with real-time updates on the events, weather, and transportation schedules, and explore new places smoothly with detailed, high-quality maps with GPS features.	High
Data Security and Privacy	Protect user data with strong security measures and comply with privacy standards, as well as integrating authentication methods, giving users confidence that their information is safe.	High
Transportation and Booking Options	To make the users' travel arrangements easier, the app provides comprehensive information and booking options for various transportation methods, including public transit, car rentals, and taxis. Additionally, it enable users to explore and book various types of accommodations, from hotels to vacation rentals, directly through the platform, making it convenient to secure lodging that fits their needs.	Medium
Multi-Platform Compatibility	Ensure the app works seamlessly across different devices and operating systems, providing a consistent and reliable experience no matter what device the user is on.	Medium
Offline Functionality	It offers offline access to crucial features like maps and saved calendars, ensuring seamless usability even without an internet connection.	Medium
User Account Management	It includes features for easy account creation, authentication, login, and profile management, allowing users to save travel preferences and itineraries.	Low
Feedback and Support	Offer a way for users to share their feedback and access help, ensuring they have the support they need while using the app.	Low

2.2. User and System Requirements:

Destination Information	
User Requirements	1. The user should have the option to explore insights on popular tourist spots in Saudi Arabia with ease and convenience. 2. The user should have access to information on various attractions like historical sites, museums and natural wonders in the region. 3. The user should be able to filter destinations based on categories such as, historical significance, natural beauty or museum exhibits.
System Requirements	1. The system must present destination details including written descriptions, images and user ratings. 2. The system must be able to refine the search outcomes by location, attraction type or overall popularity. 3. The system should be able to update destination details frequently using third party APIs or manual input sources.

Itinerary Planning	
User Requirements	1. The user should be able to customize travel plans based on their preferences, schedules and interests. 2. The user should be able to filter destinations from their travel itinerary. 3. The user should be able to save and make edits to their itineraries for future use.
System Requirements	1. The system must offer a user interface, for creating, saving and updating travel plans. 2. The system must save user preferences like travel time and preferred attractions to give recommendations tailored to them. 3. The system should include integration with calendars and options for users to set reminders or receive notifications for planned visits.

GPS and Real-time Updates

User Requirements	1. Users should be able to explore new places confidently with accurate directions and navigation. 2. Users should be able to receive and have the option to filter real-time notifications about events, weather, and transportation schedules. 3. Users should be able to see these updates without needing to refresh the app.
System Requirements	1. Notifications should appear as push alerts or in-app messages without users having to manually check. 2. The system must pull real-time data from APIs for events, weather, and transport info and show it to the user automatically. 3. The system should let users set preferences on what type of updates they want, like only getting transport info or weather alerts. 4. The system should allow offline access to cached maps and saved routes for users without an internet connection. 5. The system must integrate with a map service (like Google Maps) to provide real-time location tracking using GPS.

Data Security and Privacy

User Requirements	1. The user's personal information should be protected with strong security measures. 2. Users should be able to access their account securely using passwords or other security features like two-factor authentication. 3. The user should have control over their data and the ability to delete their account.
System Requirements	1. The system must allow users to adjust privacy settings and manage their data, including the ability to delete their account and information. 2. The system must support secure authentication methods, such as passwords and optional two-factor authentication (2FA). 3. The system must comply with privacy standards like GDPR, give users control over their data and privacy preferences, and encrypt sensitive user data during storage and sharing. 4. The system should include factor authentication, for user accounts.

Transportation and Booking Options

User Requirements	1. Users should be able to compare different transport methods, check prices, and schedules. 2. Users should be able to search for different types of places to stay, like hotels or vacation rentals. 3. Users should be able to book a ride directly through the app, like reserving a taxi or renting a car. 4. Users should be able to book their accommodation directly through the app, with a secure payment option.
System Requirements	1. The system must connect to APIs for public transit, car rentals, and taxi services to provide real-time availability and pricing. 2. The system should allow users to filter and sort accommodations based on their preferences, such as price, location, and amenities. 3. The system must support secure online booking and payment for transport services through the app. 4. The system must integrate with third-party services or APIs for accommodations to provide available options, pricing, and descriptions.

Multi-Platform Compatibility

User Requirements	1. The user should be able to use the application with different operating systems, like iOS and Android as well as devices such as smartphones, tablets or desktops.
System Requirements	1. The system must be compatible with different devices and operating systems to give users a seamless experience no matter which platform they use. 2. The system should support different screen sizes and device types using responsive design and adaptive layouts. 3. The system must undergo testing on a wide range of devices to achieve optimal performance.

Offline Functionality

User Requirements	1. The user should be able to access functions, like maps and saved travel plans even when offline.
System Requirements	1. The system must store data (such as maps and itineraries) on the device for access without an internet connection. 2. The system should automatically update stored data when the user is online to keep the information current. 3. Offline functionality must include a lightweight version of necessary services to save device storage space.

User Account Management

User Requirements	1. The user should be able to create, log in and manage their account. 2. The user should be able to save account settings, travel preferences and itineraries.
System Requirements	1. The system must provide user authentication (login, registration) and session management. 2. The system must allow users to store and retrieve personal data (Ex: saved itineraries, preferences) securely. 3. The system should offer features for users to manage account settings (Ex: password changes, privacy settings).

Feedback and Support

User Requirements	1. The user should be able to provide feedback and access support through the app.
System Requirements	1. The system must include a feedback form or portal for users to submit issues, suggestions, or complaints. 2. The system must provide a help and support section with FAQs, guides, or live support options.

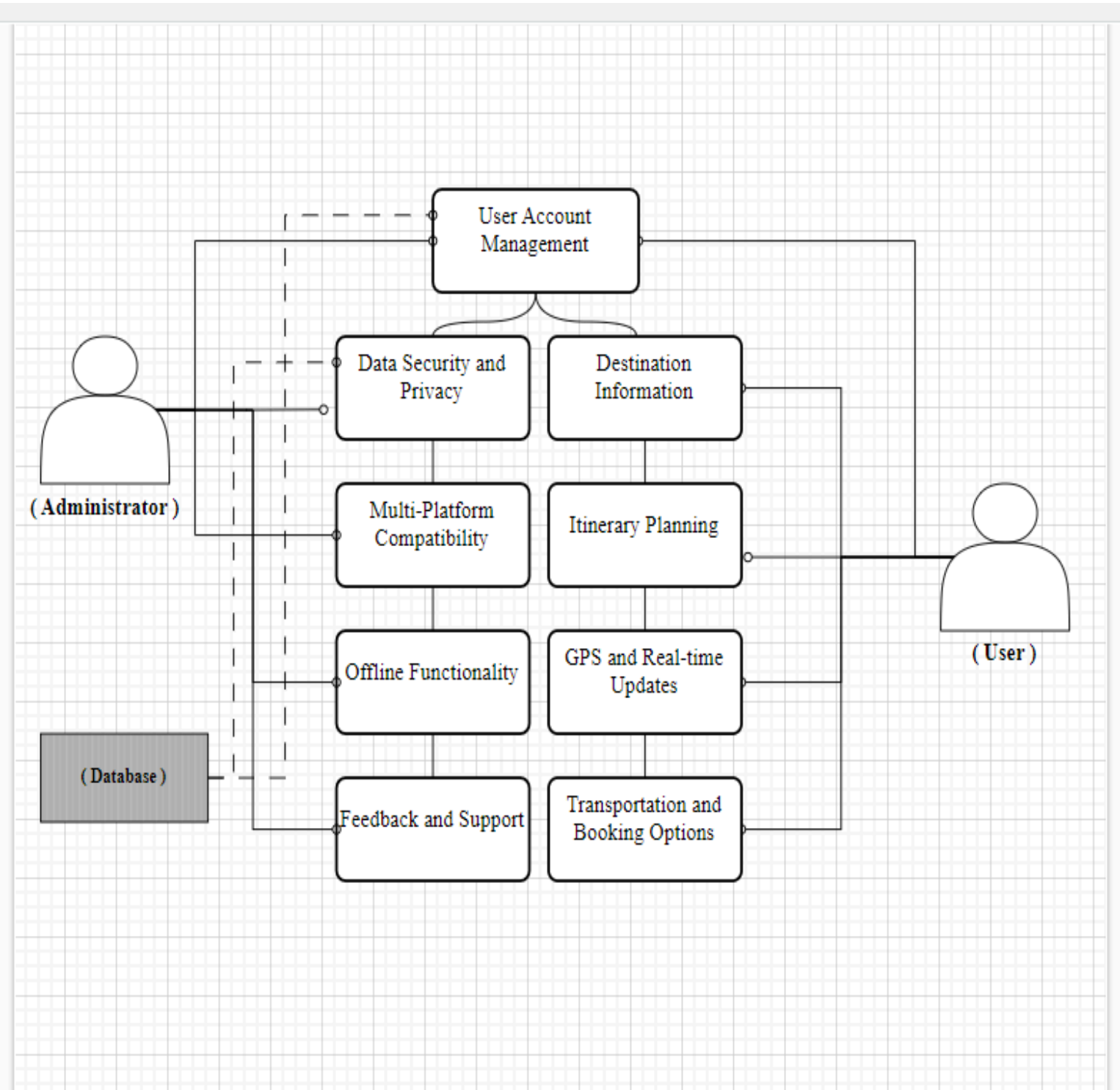
3. Sprint planning

3.1 Sprint backlog :

Backlog item		Tasks	Estimated Time
1	offering detailed insights into top tourist attractions, historical sites, museums, and natural wonders in Saudi Arabia for a comprehensive travel guide.	● Create a layout with sections for attractions, history, and nature. ● Add detailed descriptions, images, and maps for each site.	8h
2	Providing tools to help users create personalized travel plans based on their preferences, schedule, and interests for a unique, easy, and organized trip.	● Design an easy-to-use interface for users to create their personalized travel plans. ● Validate the accuracy and usability of travel plan customization and scheduling features.	7-8h
3	Users can be informed with real-time updates on the events, weather, and transportation schedules, and explore new places smoothly with detailed, high-quality maps with GPS features.	● Design a system for integrating real-time updates for events, weather, and transportation schedules, along with detailed maps and GPS features. ● Ensure that the real-time updates, maps, and GPS features work correctly and provide a smooth user experience.	8-9h
4	Protect user data with strong security measures and comply with privacy standards, as well as integrating authentication methods, giving users confidence that their information is safe.	● Develop a comprehensive security plan that includes data encryption, secure authentication, and privacy controls. ● Test the website under various scenarios to verify that user data is securely protected and that security features function correctly.	8h

5	To make the users' travel arrangements easier, the app provides comprehensive information and booking options for various transportation methods, including public transit, car rentals, and taxis. Additionally, it enable users to explore and book various types of accommodations, from hotels to vacation rentals, directly through the platform, making it convenient to secure lodging that fits their needs.	● Create a system that search and filter options, booking forms, and confirmation screens for transportation and accommodations. ● Design a confirmation screen that displays all relevant details of the booking, including transportation and accommodation specifics, booking reference, and payment information.	5h
6	Ensure the app works seamlessly across different devices and operating systems, providing a consistent and reliable experience no matter what device the user is on.	● Create a design that is flexible and adaptable to various screen sizes and operating systems. ● Test on a variety of devices and operating systems to identify any issues with compatibility or performance.	5h
7	It offers offline access to crucial features like maps and saved calendars, ensuring seamless usability even without an internet connection.	● Design offline functionality, ensuring maps are pre-loaded and saved content is easily accessible. ● Ensure that the web can seamlessly switch between online and offline modes.	5h
8	Include features for easy account creation, login, and profile management, allowing users to save their travel preferences and itineraries.	● Create a user-friendly interface for account creation, login, and profile management. ● Conduct thorough testing of the account creation, login, and profile management features to ensure they work	2-3h
9	Offer a way for users to share their feedback and access help, ensuring they have the support they need while using the app.	● Create a straightforward and accessible feedback and help section within the web. ● Ensure there's an easily navigable help section with FAQs, contact options, and support resources.	2-3h

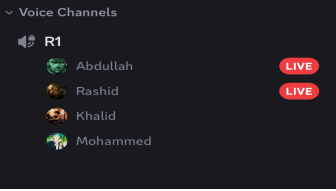
3.2. Use Case:



3.3 Group meeting report:

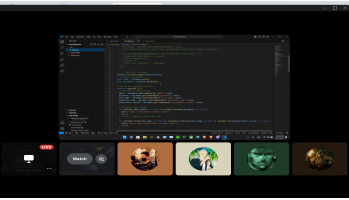
Date: 8/9/2024

Done writing the project description.



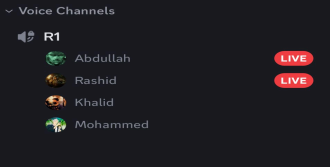
Date: 10/9/2024

Done listing most of the functionalities.



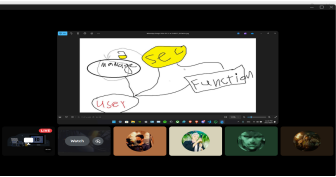
Date: 11/9/2024

Done writing the users stories and prioritizing them.



Date: 13/9/2024

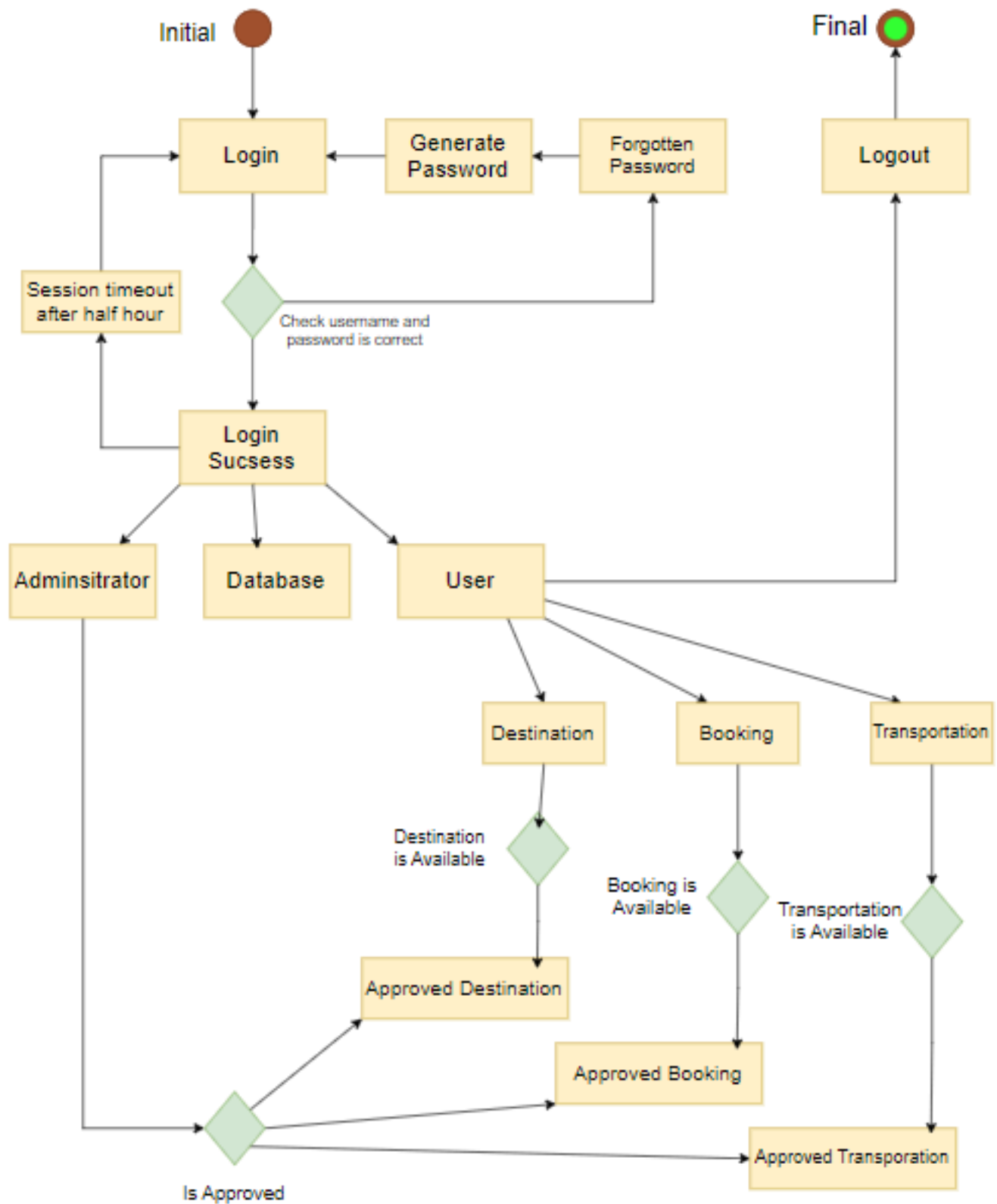
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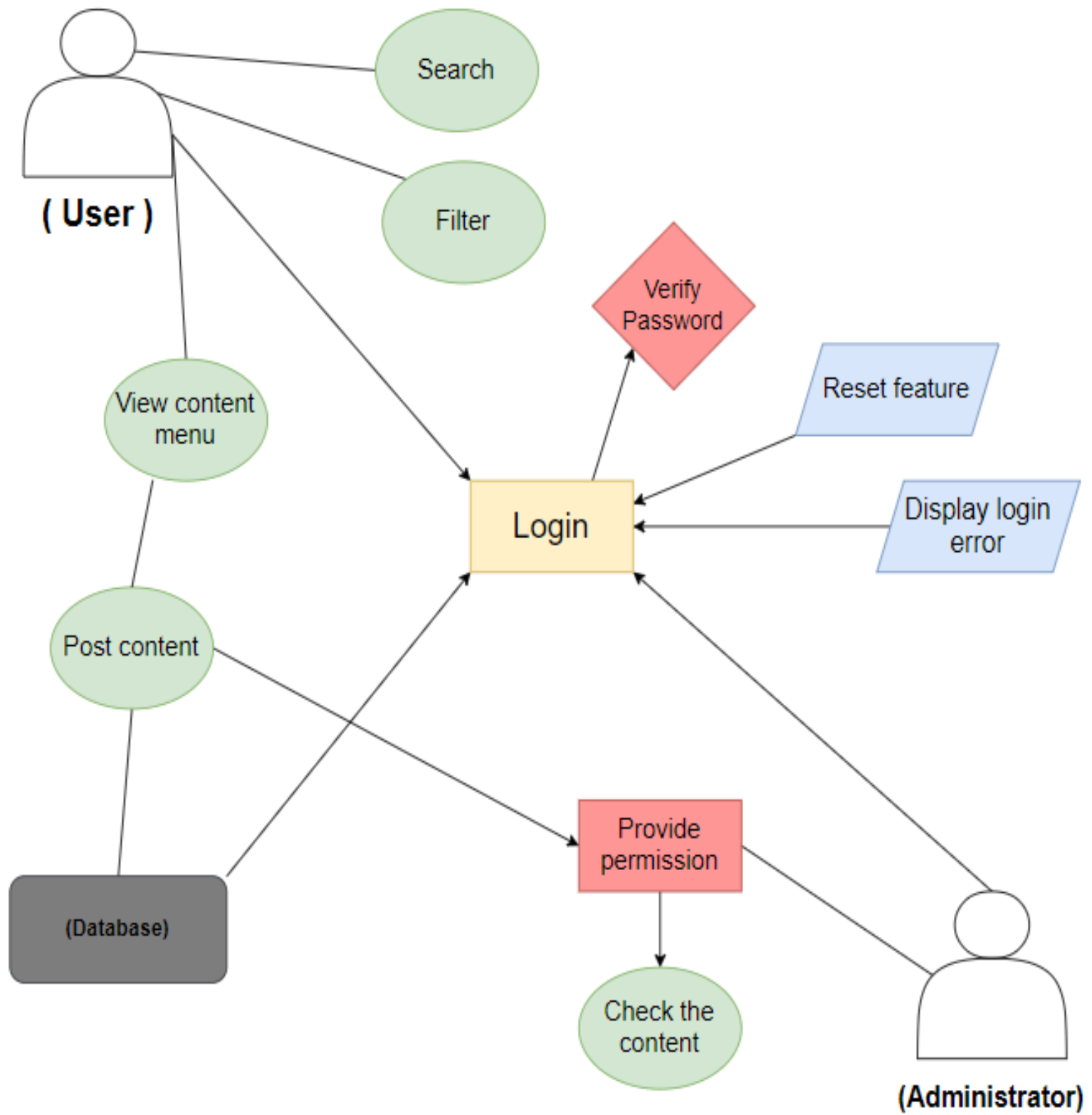
Phase Two

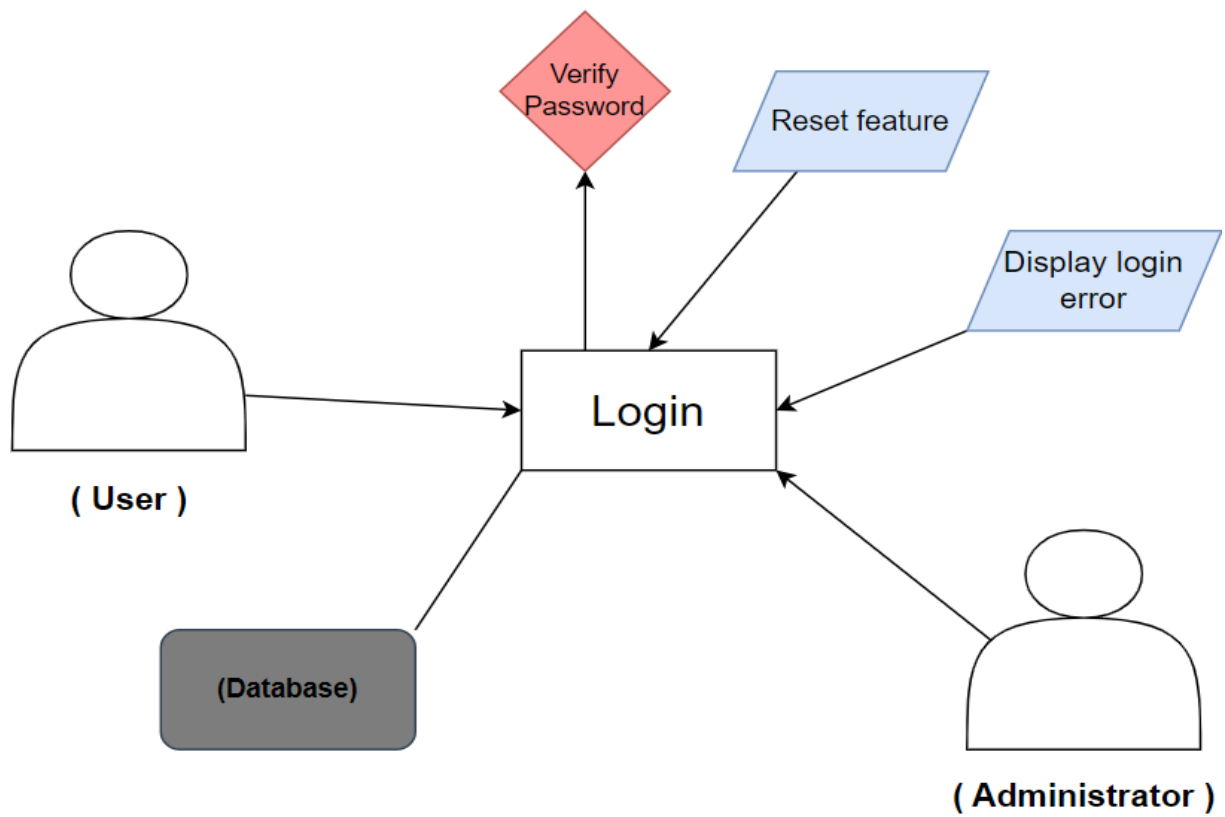
4. Design the system model

4.1 Activity diagram



4.2 Use case diagram



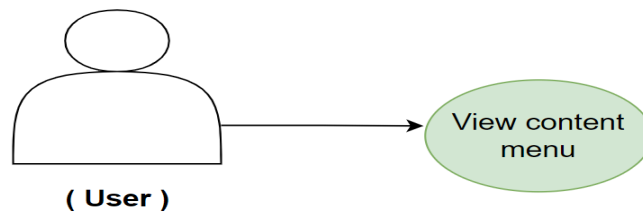


System	Tourist Guide System.
Use Case	Log in
Actors	User, Admin.
Destination	Access to accounts.
Description	The users enter their email and password, the system will check the provided information, if it is correct, this allows the user to log in.
Data	User email and password.
Pre-Condition	User must exist in the database. User must open the website.
Post-Condition	User will be logged in to the system.

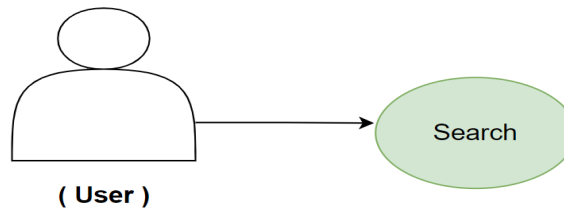
System	Tourist Guide System.
Use Case	Verify password.
Actors	User, Admin.
Destination	To ensure it is not unauthorized access.
Description	The system will check the password that the user entered with database.
Data	The entered password, the password in the database.
Pre-Condition	The user must have an account in the database. The user must enter the password.
Post-Condition	The password has been checked.

System	Tourist Guide System.
Use Case	Display Login Error.
Actors	User, Admin.
Destination	User can know there is an error entered.
Description	Users enter the wrong password or email, the system will display a message error, then the user can try again with correct information.
Data	The entered password, the password in the database.
Pre-Condition	The user must have an account in the database. The user must enter the password. The system verifies the password.
Post-Condition	Log in error message has been displayed.

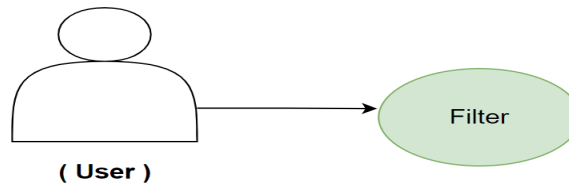
System	Tourist Guide System.
Use Case	Reset feature.
Actors	User, Admin.
Destination	User can receive an email with his password.
Description	The user will enter the password, if it is wrong, he can choose the forget button to receive an email or SMS to generate a new password.
Data	The information about the user, his email, and password in database.
Pre-Condition	The user must have an account in the database. The user must enter the password. The user must click on forget password.
Post-Condition	User password has been sent to the email.



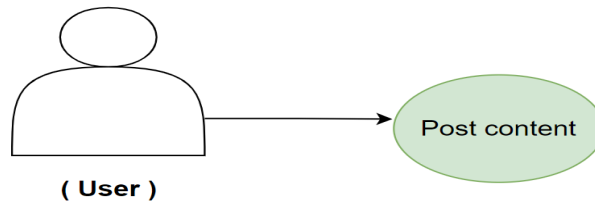
System	Tourist Guide System.
Use Case	View content menu.
Actors	User.
Destination	To choose a specific content.
Description	The system will display the content's menu and the user will choose one.
Data	Content menu components such as Itinerary Planning, destination and transportation.
Pre-Condition	Content menu must already exist in the system.
Post-Condition	Content menu has been displayed.



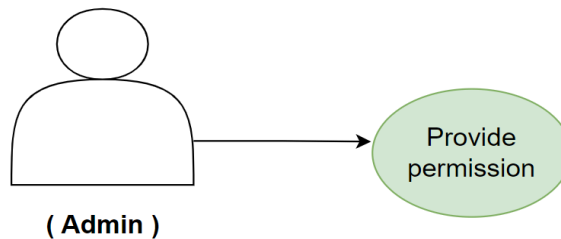
System	Tourist Guide System.
Use Case	Search.
Actors	User.
Destination	To find a specific content and any word inside the content easily.
Description	The user can search for a specific category and look for a specific content.
Data	The title of the content, The entered category.
Pre-Condition	The content must exist in the database. The category must have existed in the content menu.
Post-Condition	Content has been displayed.



System	Tourist Guide System.
Use Case	Filter.
Actors	User.
Destination	Find a specific category and content.
Description	The user can use a filter to find a specific content.
Data	Itinerary Planning, destination and transportation.
Pre-Condition	The content must exist in the database.
Post-Condition	Content has been displayed.

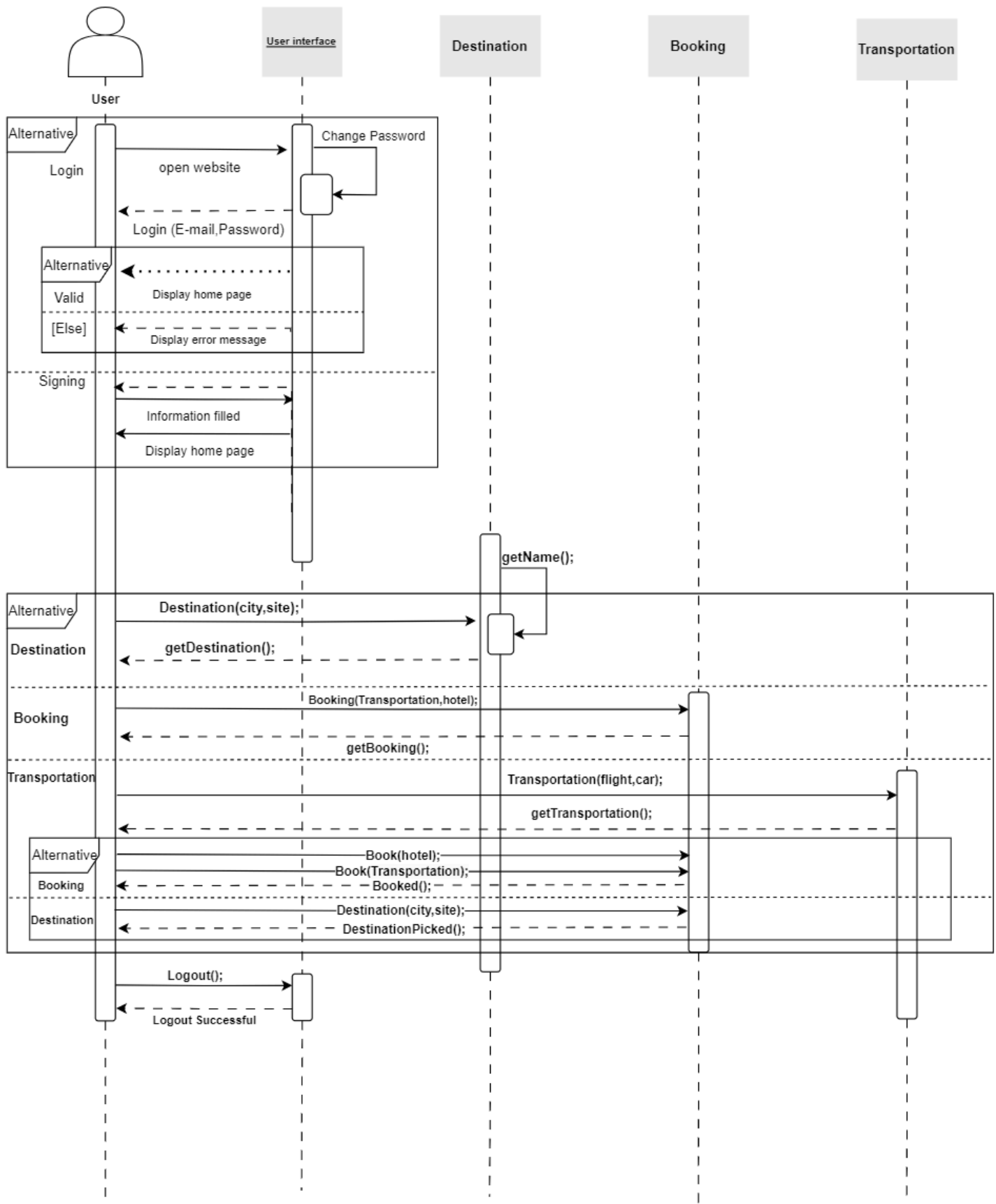


System	Tourist Guide System.
Use Case	Post content.
Actors	Publisher (User).
Destination	To make users have more spots to explore and more detailed information on the website.
Description	the publisher could post content in a specific category for further information.
Data	Information about tourism.
Pre-Condition	The content must be checked by admins before posting.
Post-Condition	The content is posted.

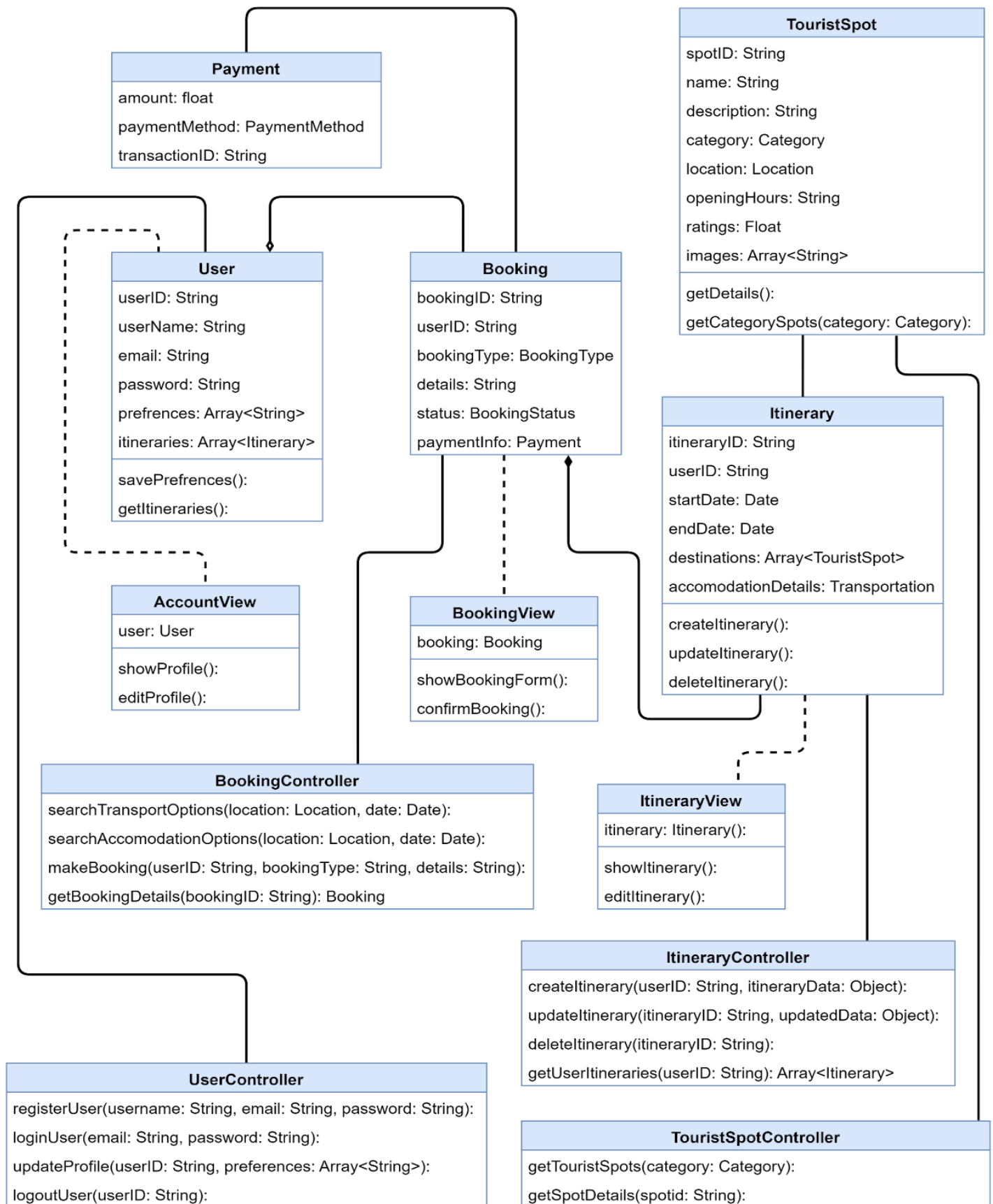


System	Tourist Guide System.
Use Case	Provide permission.
Actors	Administrator.
Destination	To let the admin get a notification about any post content request.
Description	The publisher asks permission from the admin, and they will check the content if it is valid.
Data	Information about the publisher's previous content, information about the publisher, content post request.
Pre-Condition	The publisher must send a permission message to the admin.
Post-Condition	The content is valid.

4.3 Sequence Diagram



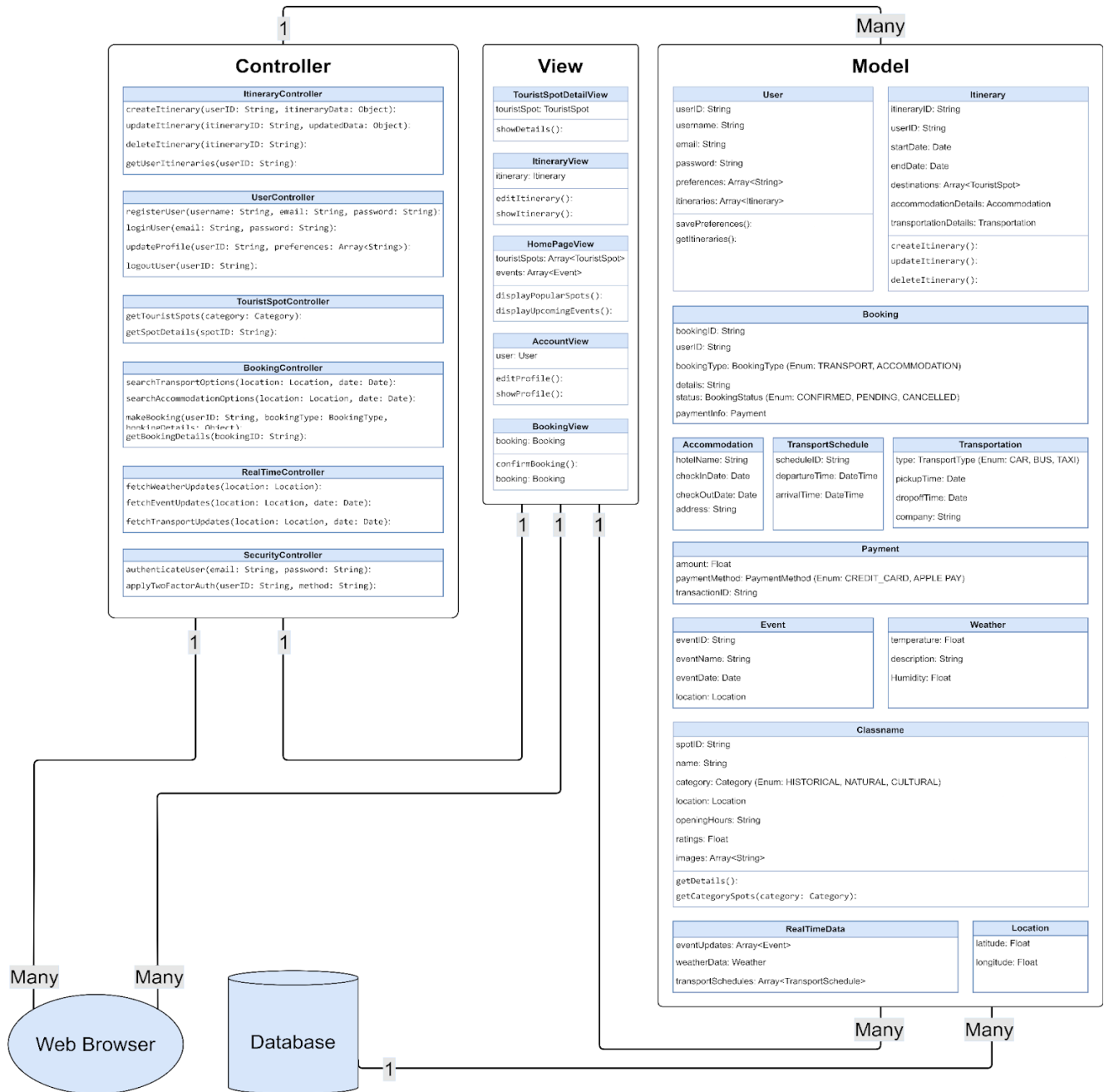
4.4 Class Diagram



5. Architectural Design and Pattern

5.1 MVC pattern diagram:

To view the diagram in full quality please [CLICK HERE](#).



The **MVC** design is the best choice for this project because it keeps everything organized by separating the **Model**, **View**, and **Controller**. This makes it way easier to maintain and update, especially when the project gets bigger or more complex. Since each part is independent, you can easily update the UI without messing up the data or logic. Plus, it's perfect for adding new features without breaking other parts of the app. Also, MVC makes testing simple, speeds up development, and keeps the code clean. It's just way more flexible and helps avoid a lot of headaches later.

Phase Three

6. Test Cases:

6.1. Log-in

ID	Test cases	Pre-condition	Test steps	Data	Expected result	Actual result
T1-Login	Successful login	Email and password need match to login	1- Enter a valid Email in the Email field 2- Enter a valid password in the password field 3- click on the "Login" button	A@gmail.com 1q2w3e4r	User should be successfully logged in and the home page should appear	work
T2-Login	Unsuccessful login with wrong inputs	Email and password need to match to login	1-enter wrong email or wrong password 2- click on the "Login" button	A@.com 1q2wdfs2	Unsuccessful login and error message appear	fail

6.2. Subjects menu

ID	Test cases	Pre-condition	Test steps	Data	Expected result	Actual result
T4-	Successful found a list of subjects	1-The user has a valid account and is logged into the system. 2- The subject's menu option is accessible and functional.	1-open the website. 2-click on the subject's menu option to access the subjects list. 3-click on a specific subject.	No specific data is used for this test case.	The system should successfully display a list of subjects	work
T5-	Successful found a list of subjects	1-The user don't need a valid account and is logged into the system. 2- The subject's menu	1-open the website. 2-click on the subject's menu option to access the	No specific data is used for this test case.	The system should successfully display a list of subjects	work

		option is accessible and functional.	subjects list. 3-click on a specific subject.			
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6.3. Search bar

ID	Test cases	Pre-condition	Test steps	Data	Expected result	Actual result
T6-search	Successful search-matah found - A user can find specific subjects by typing related keywords	Search on specific data	1-type any keyword related to a subject 2-press enter button	Database	The system needs to show the subjects that correspond to the keyword entered.	fail

T7-search	Successful search-matah found - A user can't find specific subjects by typing not related keywords	Search on specific data	1-type any keyword that doesn't match any subject 2-press enter button	Database	The system should show a "No subject found" message if no subjects match.	fail
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6.4. Ease of use

ID	Test cases	Pre-condition	Test steps	Data	Expected result	Actual result
T8-Esae of use	Successfully navigated	1 - The user must be logged into their account. 2 - The user must have access to the booking page.	1 - Open the website 2 - Navigate to the booking page 3 - Click the back or forward navigation button	NO DATA	Navigate must be working	work

T9- Esae of use	Navigation failed due to not using the button.	1 - The user must be logged into their account. 2 - The user must have access to the booking page.	1 - Open the website 2 - Navigate to the booking page 3 - don't Click the back or forward navigation button	NO DATA	Navigate not working with the web structure	fail
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6.5. Interface viewing

ID	Test cases	Pre-condition	Test steps	Data	Expected result	Actual result
T11- interf ace viewing	A user-friendly interface was successfully implemented.	The website is accessible and compatible with a wide range of devices and operating systems	1-Open the website on a desktop device 2-Verify that it adapts to the different screen sizes, remaining visually appealing and easy to use.	NO DATA	A user-friendly interface was successfully achieved	work

T12- interf ace viewing	The user-friendl y interface was not successfully implemente d.	The website is designed to be accessible and compatible across various devices and operating systems.	1-Open the website on various devices 2-Verify that it does not adapt to different screen sizes and fails to remain visually appealing or easy to use.	NO DATA	A user-friendly interface was successfully achieved	fail
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6.6. Booking

ID	Test cases	Pre- condition	Test steps	Data	Expected result	Actual result
T13- booking	Successful Booking Process.	1 - The user must be logged into their account. 2 - The booking slots or services must be available for the chosen date and time.	1-Open the website and navigate to the booking page. 2-Submit the booking request.	NO DATA	The booking page loads correctly, and the user is prompted to log in if not already authenticated.	work

T14-booking	Ensure the system appropriately handles errors during the booking process and displays meaningful feedback to the user.	1 - The user must be logged into their account. 2 - The booking slots or services must be available for the chosen date and time.	1-Open the website and navigate to the booking page. 2-Submit the booking request.	NO DATA	The user is unable to complete the booking	fail
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6.7. Payment method

ID	Test cases	Pre-condition	Test steps	Data	Expected result	Actual result
T15-Payment method	Successful Payment Process.	1 - The user must be logged into their account. 2 - The user has successfully selected a service or event to book 3 - A valid payment method is	1-After completing the booking details, proceed to the payment section. 2-Enter valid payment details (card number, expiry date, CVV).	NO DATA	The system processes the payment successfully and confirms the booking.	work

		available (e.g., credit/debit card, PayPal).				
T16- Payment method	Unsuccessful Payment Process.	1 - The user must be logged into their account. 2 - The user attempts to use an invalid payment method (e.g., expired card, insufficient funds, or wrong details).	1-After completing the booking details, proceed to the payment section 2-Select a payment method that is likely to fail (e.g., an expired card, invalid card details).	NO DATA	The system fails to process the payment and displays a clear error message (e.g., "Payment failed," "Card declined," or "Insufficient funds").	fail